

# GERTIAN MICI

Senior Software Engineer — .NET, Distributed Systems & Cloud

+46 729 616 485 | gertian.mici@gmail.com | Stockholm, Sweden (open to remote)

linkedin.com/in/gertian-mici-2102 | github.com/GertianMici | gertianmici.com

English (C1) | Albanian (native)

## PROFESSIONAL SUMMARY

---

Senior software engineer with 7 years building reliable, cloud-native systems in .NET — from real-time applications and single sign-on to a large-scale navigation platform and, today, core banking for a consumer fintech app. Deep experience in microservices, event-driven architecture and a hand-written data access layer with no ORM, where the schema, indexes, stored procedures and versioned migrations are mine. Comfortable across the stack when the problem needs it. I verify before I claim, and I fix the whole class of a bug rather than the single instance reported.

## TECHNICAL SKILLS

---

**Languages:** C#, T-SQL, Python, TypeScript, JavaScript

**Backend & Architecture:** .NET 8/7/6/Core, ASP.NET Core Web API, microservices, event-driven architecture, CQRS, clean architecture, REST and gRPC APIs (Protobuf), SignalR, domain-driven design

**Distributed Systems:** RabbitMQ, Kafka, transactional outbox, idempotency and retry design, distributed locking, retryable-vs-permanent failure handling, rate-limit handling, distributed tracing

**Data:** SQL Server, Azure SQL, PostgreSQL, MongoDB, Redis, schema and index design, stored procedures, versioned migrations, query and index optimisation, BigQuery analytics SQL

**Cloud & DevOps:** Azure (SQL Database, Blob Storage, Key Vault, AKS), Docker, Kubernetes, Argo CD / GitOps, GitHub Actions

**Security:** PCI-scoped development, applied cryptography, key management and rotation, OAuth2, IdentityServer4 (PKCE), secrets handling, least-privilege data access, PII masking

**Observability & Testing:** Elasticsearch, Kibana, correlation-id tracing, structured logging, MSTest, xUnit, FluentAssertions, integration and cross-service regression testing, BenchmarkDotNet

**Developer Tools:** Git, GitHub Actions, Claude Code, Cursor, GitHub Copilot

## PROFESSIONAL EXPERIENCE

---

### Senior Software Engineer, MAJORITY

Stockholm, Sweden | Jan 2024 – Present | Core banking for a consumer fintech app

- **Core banking platform.** Accounts, ledgers and transaction posting for a consumer fintech app: how each card transaction is authorized, cleared, settled and posted, and the services that keep balances and ledger entries in agreement.
- **Settlement and ledger correctness.** Resolving discrepancies found in production and closing each one with MSTest regression tests so it cannot reappear.
- **Failure classification across services.** Separated permanent failures from retryable ones so callers stop retrying requests that can never succeed, and rolled it out without changing the behaviour of any existing handler on deploy. Own the messaging underneath: RabbitMQ handlers, the transactional outbox, and idempotency keys on retryable work.
- **New credit service and crypto-backed wallet.** One of the first engineers on a consumer credit service, from an empty repository to live account creation and card issuing within 2 quarters. Backend for a crypto-backed global wallet with a staged rollout behind feature flags, including Solana USDC transactions via Privy wallet.
- **Cardholder-data cryptography, PCI scope.** Card fingerprinting, sensitive credentials designed so they are never stored, and key rotation carried through to retiring the previous scheme. Consolidated duplicated hashing logic into a shared platform library and migrated 7 downstream services onto it in one release.
- **Production incident investigation.** Daily in Elasticsearch and Kibana — error triage, correlation-id traces and call-tree reconstruction across service hops.
- **Platform.** A .NET major-version upgrade across 6 backend services in 2 weeks and a move to Kubernetes-native GitOps delivery, each service shipping as a Docker container onto AKS, over Azure SQL Database, Blob Storage and Key Vault.

### Senior Software Engineer (Contract), Qbrick

Stockholm, Sweden | Sep 2023 – Jan 2024

- Stabilised event-driven services: resolved memory allocation issues and eliminated throughput bottlenecks under sustained load.
- Reduced database read latency by roughly 50% through index and query optimisation.
- Strengthened CI/CD pipelines and health checks to improve release confidence.

### Software Engineer / Team Lead, Lufthansa Industry Solutions

Tirana, Albania | Jun 2022 – Sep 2023

- Led backend development for a large-scale navigation platform covering routing, rewards, favourites and vehicle domains.

- Built .NET microservices communicating over gRPC and Kafka, with Redis, PostgreSQL and MongoDB for persistence and caching, and xUnit suites covering the service layer.
- Supported onboarding of 70,000+ users within 48 hours with no customer-impacting incidents; defined reliability targets and the dashboards to track them.

### **Software Developer, S2 Global (an OSI Systems company)**

*Tirana, Albania | Mar 2021 – Jun 2022*

- Developed high-availability cloud services and implemented single sign-on with IdentityServer4 (Authorization Code with PKCE).
- Worked alongside an Angular team on UI integration requirements and authentication flows.

### **Full-stack Software Developer, Soft & Solution**

*Tirana, Albania | Sep 2019 – Feb 2021*

- Delivered .NET MVC and SignalR applications plus a Xamarin mobile application across small, fast-moving teams.
- Built real-time features and maintained SQL Server performance for transactional workloads.

## **SIDE PROJECTS**

---

- **Helkost — offline food-processing scanner for iOS ([gertianmici.com/helkost](https://gertianmici.com/helkost))**. Swift, SwiftUI, SwiftData, CloudKit, HealthKit. Ships 1.1M products inside the app, so a barcode resolves with no network. The build pipeline cuts a 1.2 GB export to the rows carrying a processing verdict and writes them to a WITHOUT ROWID table in primary-key order, so the table is its own index and no page splits happen. Barcodes outside it fall back to the live API.
- **Osrever — iOS reverse-audio party game ([gertianmici.com/osrever](https://gertianmici.com/osrever))**. SwiftUI, AVFoundation. One player records a word; the app reverses the PCM samples for a second player to mimic, then reverses their attempt to reveal how close they got. Entirely on-device: no networking, no analytics, no third-party SDKs, and a privacy manifest declaring zero data collection. Ships via a GitHub Actions pipeline that signs with an App Store Connect API key and promotes a release branch to TestFlight.
- **Everyday — language-immersion game for iOS**. Swift, SwiftUI, structured model output. Every character is a language model held to the learner's CEFR level, with a goal to reach in the target language. Bundled content stores the intent of an exchange rather than any target-language text, so adding a language is configuration rather than a translation project — enforced by a test. The whole turn (line, translation, emotion, goal state, new vocabulary) returns from one call against a strict JSON schema, so decoding has no optionals and no retry path.
- **ClipClap — macOS clipboard manager ([github.com/GertianMici/ClipClap](https://github.com/GertianMici/ClipClap))**. Swift, AppKit. Windows-style clipboard history for macOS: recall with a global hotkey, search the history, and paste by number key. Open source.
- **Browser games — nine games, no engine ([gertianmici.com/#/games](https://gertianmici.com/#/games))**. Three.js and Canvas. 3D racing and chess, a maze arcade game, a cellular-automaton sand sandbox, a one-button stacker and two idle games, each written from scratch with sound synthesised at runtime.

## **EDUCATION & CERTIFICATIONS**

---

### **BSc, Computer Engineering, Polytechnic University of Tirana**

*Tirana, Albania | Oct 2017 – Aug 2020*

Microsoft Certified: Azure Fundamentals (2023) · Microsoft Exam 480: HTML5 with JavaScript and CSS3 (2020)